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Offline Voice-Driven Advisory Checklist Reader Using a Galvanically-Isolated Receive-Only Audio-Panel Tap and a Dedicated Galvanically-Isolated Audio-Panel Output Channel, with a Dual-Removable-Card Fail-Safe Configuration Architecture

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PROVISIONAL APPLICATION FOR PATENT UNDER 37 C.F.R. § 1.53(c)

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Title of Invention: Offline Voice-Driven Advisory Checklist Reader Using a Galvanically-Isolated Receive-Only Audio-Panel Tap and a Dedicated Galvanically-Isolated Audio-Panel Output Channel, with a Dual-Removable-Card Fail-Safe Configuration Architecture

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Cross-Reference to Related Applications

None. This is the first filing for this subject matter and is intended to establish a priority date. A nonprovisional and/or PCT application claiming benefit of this provisional may be filed within twelve (12) months.

Field of the Invention

The invention relates to flight-deck advisory and human-factors equipment, and more particularly to a standalone, offline electronic device that audibly reads aircraft checklist items and advances through them in response to recognized crew speech, wherein the crew speech is obtained from a passive, receive-only, galvanically-isolated INPUT tap of an aircraft audio panel, and wherein checklist read-aloud audio is delivered to the crew in-headset through a separate, dedicated, galvanically-isolated OUTPUT channel into the audio panel (a "COM3-style" auxiliary channel), and wherein all aircraft-specific behavior is supplied by removable media using a fail-safe dual-card configuration architecture. The audio-panel INPUT tap and the audio-panel OUTPUT channel are on separate galvanically-isolated paths; the apparatus includes no onboard speaker in the installed configuration.

Background

Modern flight decks use both paper Quick Reference Handbooks (QRHs) and, in larger aircraft, integrated Electronic Checklist (ECL) systems. Integrated ECLs (e.g., systems embodied in Boeing, Honeywell, and Collins Aerospace avionics suites) are tightly coupled to the host aircraft's certified avionics and crew alerting systems. As a result they are expensive, aircraft-type-specific, require substantial software assurance (e.g., RTCA/DO-178C), and are impractical to add to the large installed base of general-aviation and light-business aircraft that still rely on paper checklists.

Voice-interactive checklist concepts are known in principle. Prior systems read checklist items aloud and advance on a spoken acknowledgment word, but they share one or more of the following limitations:

1. They depend on integration with the host avionics suite, on a dedicated onboard microphone, or on cloud / large-vocabulary speech-to-text services, making them connectivity-dependent, non-deterministic, costly to certify, or all three.
2. They use a dedicated microphone that re-captures cabin/cockpit acoustics — degraded by ambient noise, oxygen masks, and headset use — rather than the clean, already-mixed audio the crew actually hears.

3. They tie checklist content to firmware, so supporting a new aircraft type requires a software change and re-verification.
4. They lack a rigorous, testable fail-safe behavior that guarantees the crew is never shown partial, stale, or mismatched checklist content.

A device that (a) is electrically and functionally minimal, (b) attaches to the aircraft only through a one-way, isolated audio tap, (c) is fully offline and deterministic, and (d) is rendered aircraft-specific entirely by removable media would be far cheaper to build and substantially easier to qualify (e.g., suitable for a Non-Required Safety Enhancing Equipment, "NORSEE," approval path emphasizing environmental qualification rather than complex software assurance). No known device combines these attributes. The present invention addresses this need.

Summary of the Invention

Disclosed is a standalone, offline electronic apparatus and corresponding method for advisory reading of aircraft checklists, comprising, in various aspects:

- An audio-input stage that derives crew speech from a passive, receive-only tap of an aircraft audio panel. The INPUT tap is galvanically isolated from the aircraft (for example, by a line-level isolation transformer in an analog embodiment, or by a buffered, receive-only digital feed in a digital embodiment) such that via this INPUT path the apparatus cannot transmit on, key, mute, inject into, or back-feed the audio panel, and a fault, short, open, or loss of power within the apparatus cannot load down or alter audio-panel function through this path. The tapped, already-mixed crew audio — rather than a dedicated onboard microphone — is the primary input to the speech recognizer.
- A dedicated, galvanically-isolated audio OUTPUT stage that injects checklist read-aloud audio into a separate, dedicated aircraft audio-panel channel (a "COM3-type" auxiliary audio channel). The OUTPUT stage (for example, an I2S DAC followed by a line-level isolation transformer on the TX output line) delivers checklist items to the crew in their headsets, replacing the apparatus's own onboard speaker. The OUTPUT channel is on a separate, galvanically-isolated path and a separate connector from the receive-only INPUT tap, such that a fault, short, open, power loss, or output-stage failure within the apparatus cannot key, jam, load, or back-feed the audio panel's other channels or required COM radios; the injected audio is advisory/informational only on one dedicated aux channel and does not constitute a keying, command, or control interface to the aircraft. This dual-isolated-channel architecture — a receive-only isolated INPUT tap and a separately isolated dedicated advisory-audio OUTPUT channel — may itself represent a novel combination.
- An offline, deterministic, bounded-grammar speech subsystem executing on an embedded microcontroller (e.g., a dual-core system-on-chip of the ESP32-S3 class) without any network, cloud, or wireless connectivity, comprising an acoustic front end with a voice-activity detector (VAD), a wake-word stage, and a fixed/bounded command-grammar recognizer. The bounded grammar is small (well under a few hundred commands) and is loaded from removable media, keeping recognition deterministic and within the capability of microcontroller-class hardware.
- A voice-activated (VOX) interaction mode in which the VAD gates recognition so the crew operates the device hands-free by speaking, with a push-to-talk (PTT) input retained as a manual force-listen override rather than as the primary trigger. The VOX behavior (sensitivity, hangover time, and whether PTT override is enabled) is configurable from removable media.

- A read-aloud playback subsystem that, upon recognizing a checklist-invocation phrase, plays a stored audio clip for each checklist item in sequence through the dedicated, galvanically-isolated audio-panel OUTPUT channel (the COM3-type auxiliary channel described above), so the crew hears checklist items in their headsets; advances to the next item upon recognizing a spoken completion/advance word. The apparatus's own onboard speaker is removed in the installed configuration (an onboard speaker output may optionally be retained as a bench-test-only output, clearly labeled as not for flight use).
- A dual-removable-card configuration architecture comprising two physically separate removable storage cards in two slots: a configuration card that defines the installation (an aircraft identifier, the selected audio-input source, VOX parameters, and fitted-hardware options) and is preferably write-protected/locked after installation to serve as a per-installation configuration-control record; and a data card that carries aircraft-specific checklist content (a checklist library, trigger and advance vocabulary, and read-aloud audio clips). The apparatus supports a new aircraft type by installing a new data card with no firmware change and no recompilation. Configuration authority (set by the installer) is thereby separated from content authority (authored from the aircraft flight manual), so a content revision cannot silently alter the installation's audio/VOX setup.
- A fail-safe "revert-to-unopened" rule. Before presenting any checklist, the apparatus validates both cards and requires that both are present, readable, schema-valid, internally complete (every referenced audio clip exists), and mutually consistent — specifically, that the aircraft identifier on the configuration card resolves to a matching content set on the data card. If any condition fails — a card is missing, unreadable, malformed, schema-invalid, references a missing audio clip, or the two cards disagree on the active aircraft — the apparatus refuses to present any checklist (its in-memory checklist store is held empty) and enters a clearly-annunciated fault state. The apparatus never presents partial, stale, or mismatched checklist content; the only failure mode exposed to the crew is loss of function, not misleading information.
- A dark-cockpit fault annunciation in which the apparatus displays nothing when selected-in and healthy, illuminates an amber fault legend on any fault, and, when deselected (selected-out), shows an "off" status while inhibiting the fault legend; a power-up lamp test verifies the annunciator before normal operation.

The combination of a receive-only isolated audio-panel INPUT tap as the speech-recognition source, a dedicated galvanically-isolated audio-panel OUTPUT channel for in-headset checklist delivery (the two channels being electrically separate with independent isolation barriers), an offline bounded-grammar VOX recognizer on microcontroller-class hardware, the dual-card configuration/content split with a cross-card consistency check, and the revert-to-unopened fail-safe with dark-cockpit annunciation provides an advisory checklist reader that is inexpensive, aircraft-agnostic, deterministic, and structured for a streamlined certification path. These aspects are individually and severally inventive and may be claimed alone or in combination. The dual-isolated-channel architecture (isolated receive-only INPUT tap + isolated dedicated advisory-audio OUTPUT channel, electrically separated) may itself be a novel combination.

Brief Description of the Drawings

(Figures to be supplied as informal drawings with the provisional; a system wiring diagram already exists in the project and may be adapted.)

- FIG. 1 — System block diagram: audio-panel receive tap → isolation stage → audio codec/ADC → microcontroller (AFE/VAD → wake word → bounded-grammar recognizer → playback sequencer) → isolated audio output stage → audio-panel COM3-style channel; plus the two card slots and the annunciator.
- FIG. 2 — Analog audio-input embodiment: high-impedance line tap → isolation transformer → series/anti-alias network → codec line-in.
- FIG. 3 — Digital audio-input embodiment: buffered, receive-only I²S/codec feed with clocks generated locally and no data driven toward any aircraft bus.
- FIG. 4 — Dual-card architecture: configuration card (slot 1) and data card (slot 2), their respective schemas, and the cross-card aircraft-identity consistency check.
- FIG. 5 — Boot/validation state machine implementing the revert-to-unopened rule, showing the healthy ("store OK") path and each fault status.
- FIG. 6 — Dark-cockpit annunciator state table (selected-in healthy = dark; fault = amber; selected-out = white "off," fault inhibited) and the power-up lamp test.
- FIG. 7 — Interaction flow: crew speaks checklist name → recognizer matches a trigger phrase → item clip plays → VAD/recognizer awaits an advance word → next item, with PTT as override.
- FIG. 8 — Card-integrity / authentication embodiment: a digital signature (or keyed MAC) over each card's manifest and referenced clips is verified at boot against a public key held in device firmware; a signature or checksum failure is treated as a fault condition under the revert-to-unopened rule.
- FIG. 9 — Tamper-evident installation/audit log: an append-only, hash-chained provenance record (configuration card) and revision history (data card), each entry linked to the prior entry by a cryptographic hash so any deletion or back-dating is detectable.
- FIG. 10 — Multi-aircraft (fleet) data card: a single data card carrying plural per-tail content sets; the cross-card aircraft-identity check selects the active set named by the configuration card and rejects operation if no set matches.
- FIG. 11 — Active output-injection safeguards: (a) automatic output inhibit/mute when a required COM transmit/keying condition is detected; (b) mute-on-fault of the output stage; (c) a self-test of the output isolation barrier at power-up; and (d) a watchdog that mutes the output if the playback sequencer stalls.
- FIG. 12 — "No-mask" audio arbitration: a detector senses required-radio/ATC audio on the receive tap and automatically ducks, pauses, or defers advisory read-aloud audio so it cannot overlap or mask required communications, resuming afterward.
- FIG. 13 — Crew readback verification: after reading an item, the recognizer matches an item-specific expected response (e.g., a value or state, "gear — up") and advances only on a correct readback, optionally re-prompting on a mismatch.
- FIG. 14 — Alternate isolation embodiments for either channel: (a) optical isolation, (b) capacitive isolation, and (c) transformer-coupled digital isolation, shown as interchangeable barrier implementations.

Detailed Description

1. Overall architecture

The apparatus is a self-contained advisory device that connects to an aircraft audio panel through two electrically separate, galvanically-isolated channels and otherwise draws power on its own protected rail. It comprises: an audio-input stage (receive-only isolated tap); a microcontroller running an offline speech-recognition and playback pipeline; an audio output stage (galvanically-isolated, line-level TX stage injecting advisory audio into a dedicated COM3-style audio-panel channel); two removable-card slots; and an annunciator with a select input and an optional push-to-talk input. The device is advisory and is not a required aircraft system; it does not interface with avionics, engine, or flight-control systems and takes no operational credit. The installed configuration has no onboard speaker; checklist audio reaches the crew via the audio-panel COM3-style output channel.

2. Audio-panel input stage (receive-only, galvanically isolated)

Crew speech is obtained from the aircraft audio panel rather than from an onboard microphone. The audio panel already presents the mixed audio the crew hears (intercom and received radio), at line/headphone levels typically on the order of 150–600 ohms and roughly 1–5 V RMS. Two interchangeable embodiments are disclosed and selectable per installation:

(a) Analog embodiment. A high-impedance tap is taken across a headphone/intercom/line output of the audio panel and passed through a 600-ohm-class audio isolation transformer (for example, an Allen Avionics AGL-series ground-loop isolation transformer), then through a series resistor (e.g., ~220–470 ohms) and a simple anti-alias network into the line-in of an audio codec/ADC (for example, an ES8388, ES7210, PCM1808, or CS5343). Because the tap is high-impedance and parallel, the panel sees a negligible load; because of the transformer, the device is galvanically isolated and cannot back-feed the panel.

(b) Digital embodiment. Where a codec sits closer to the source, a buffered, receive-only digital (e.g., I²S) feed enters the same codec/ADC. Bit clock, word-select, and master clock are generated by the device for the ADC only; no data line is driven back toward any aircraft bus.

In both embodiments the INPUT tap is strictly one-way (listen-only) on this path: via the INPUT stage alone, the device has no path to transmit, key a radio, mute, inject audio into, or otherwise alter the audio panel; and a short, open, power loss, or internal fault on the input side cannot load down, ground, or back-feed the panel on this path. This receive-only, isolated INPUT posture is a deliberate, testable safety property. The separate OUTPUT channel (Section 2a) injects advisory audio through its own isolated path and is described separately. An onboard MEMS microphone may optionally be provided for bench testing only and is expressly not the installed operational audio source.

2a. Audio-panel output stage (dedicated isolated COM3-style channel)

Checklist read-aloud audio is delivered to the crew in-headset via a dedicated, galvanically-isolated OUTPUT stage that injects audio into a separate, dedicated auxiliary audio-panel channel ("COM3-type" channel). In a representative embodiment, the microcontroller's I2S DAC output drives an I2S-to-analog DAC/line driver, followed by a line-level isolation transformer on the TX output line, followed by a line-level output to the audio panel's COM3-style input. The isolation transformer on the TX line provides galvanic isolation between the device and the panel's COM3 input, ensuring that a fault, short, open, power loss, or output-stage failure within the device cannot key, jam, load, or back-feed the panel's other channels or required COM radios.

The OUTPUT stage operates exclusively on the dedicated COM3-style auxiliary channel; it does not inject into, key, or otherwise affect the required COM1/COM2 channels or the crew intercom. The injected audio

is advisory/informational only and does not constitute a keying, control, or command interface to the aircraft or its communications systems.

The INPUT tap (Section 2) and the OUTPUT stage (this section) are on separate galvanically-isolated paths with separate connectors and independent isolation barriers, so there is no conductive path between the two interfaces inside the apparatus. This "dual-isolated-channel" architecture is a deliberate design choice that permits the apparatus to listen to crew audio and speak back to the crew in-headset while maintaining electrical separation from the aircraft on both paths.

3. Offline bounded-grammar speech subsystem with VOX

The microcontroller (e.g., an ESP32-S3-class dual-core SoC with PSRAM) runs an entirely on-device speech pipeline (for example, the Espressif ESP-SR framework): an acoustic front end (AFE) providing noise suppression and a voice-activity detector (VAD); a wake-word stage; and a fixed/bounded-grammar command recognizer (for example, MultiNet) operating on a small, enumerated vocabulary loaded from the data card. There is no Wi-Fi, Bluetooth, cellular, or cloud dependency; recognition is deterministic and the wireless interfaces are disabled, which is both a security and an emissions-qualification advantage.

The grammar is intentionally small (well under the recognizer's command cap, e.g., on the order of a few tens of commands) and bounded — checklist trigger phrases plus universal advance words and item-specific advance words. Keeping the workload to fixed-grammar command recognition (not open-ended transcription) is what keeps a microcontroller-class part in scope and yields a small, well-understood failure surface.

In the default VOX (voice-activated) mode, the VAD gates recognition: the device listens whenever speech is detected on the tapped audio-panel feed, tuned by configurable VAD sensitivity and hangover time. The push-to-talk (PTT) input is retained only as a manual force-listen override; holding it opens recognition regardless of VAD, and a configuration setting may make PTT the sole trigger (disabling VOX) for installations that prefer the legacy behavior. Because the recognizer operates on the audio-panel feed, it hears precisely the audio the crew exchanges, improving robustness relative to a cabin microphone.

4. Read-aloud playback and interaction flow

Upon recognizing a checklist-invocation trigger phrase (e.g., a spoken emergency name), the device retrieves the corresponding checklist from the data card and reads its items aloud, one at a time, by playing a stored audio clip per item through the dedicated, galvanically-isolated COM3-style audio-panel OUTPUT channel (Section 2a), so the crew hears checklist items in their headsets. After reading an item, the device awaits a recognized completion/advance word (either a universal advance word such as "check," "checked," "complete," or "next," or an item-specific advance word) before advancing to the next item. The interaction is hands-free in VOX mode, with PTT available as override. There is no onboard speaker in the installed configuration; an onboard speaker output may optionally be provided as a bench-test-only feature.

5. Dual-removable-card configuration architecture

Two physically separate removable cards (e.g., microSD/FAT32) occupy two slots:

Configuration card (slot 1). Carries a single installation-configuration manifest defining: a schema version; an aircraft identifier; the audio source (analog or digital); audio parameters (input gain, codec part, sample rate); VOX parameters (mode, VAD sensitivity, hangover, PTT-override flag); fitted-hardware options; and installation provenance (shop, installer, date, work order). The configuration card is preferably write-protected/locked after installation so it serves as the per-tail configuration-control record and so

configuration cannot drift in service.

Data card (slot 2). Carries aircraft-specific content: one folder per aircraft, each with a checklist manifest (aircraft identifier, title, universal advance words, and a checklist library of entries having stable identifiers, titles, trigger phrases, and ordered items) and a set of read-aloud audio clips, one per referenced item.

Because configuration (installation-controlled) and content (revision-controlled from the flight manual) are physically and logically separated, a content revision cannot silently change the installation's audio/VOX setup, and re-using a device on a different aircraft or wiring requires changing only the configuration card. Supporting a new aircraft type requires only authoring a new data card — no firmware change and no recompilation. A grammar validator enforces recognizer constraints on the data card (e.g., lowercase, single-spaced phrases; numbers spelled as words; command count under the recognizer cap; every referenced clip present), rejecting non-compliant cards.

6. Revert-to-unopened fail-safe and dark-cockpit annunciation

At boot, the device attempts a complete, valid load of both cards, reading the configuration card first (it names the aircraft and the audio source the input stage must initialize), then the matching data-card content. It returns exactly one status. The healthy status ("store OK") requires that both cards are mounted, both manifests parse and are schema-valid, every referenced audio clip is present, and the aircraft identifier is consistent across the two cards. Any of the following yields a fault and an empty in-memory checklist store, so no checklist can be presented even if requested: configuration card absent or unreadable; configuration manifest malformed or schema-invalid; data card absent or unmountable; data manifest missing, malformed, schema-invalid, or empty; a referenced audio clip absent; the cross-card aircraft identifier mismatched; or a memory-allocation failure during load. This is the literal "revert-to-unopened" rule: the device never presents partial, stale, or mismatched checklist content — the only crew-visible failure mode is loss of function.

Faults are annunciated using a dark-cockpit convention. When the device is selected-in and healthy, the annunciator shows nothing. On any fault, an amber fault legend illuminates. When the device is selected-out, a white "off" status shows and the fault legend is inhibited (a deliberately deselected device requires no crew action). A power-up lamp test illuminates the legend(s) briefly so a failed indicator cannot masquerade as a healthy/dark state. Recognition and fault monitoring are suspended while selected-out.

7. Representative hardware and power

A representative build uses an ESP32-S3-class module with PSRAM; an audio codec/ADC for the input path; an isolation transformer and input network for the analog input path; an I2S DAC / line driver followed by a second isolation transformer for the COM3 audio output path; two microSD slots; and a split-legend annunciator with low-side lamp drivers for the white "off" and amber "fault" legends. The device runs from a low-current supply (e.g., USB 5 V $\geq$ 1 A) on its own protected rail; neither the input tap nor the isolated output stage draws significant power from the aircraft; both are isolated from the device's grounds through their respective transformers.

8. Certification posture (advisory, non-required)

The device is advisory and non-required, fails to a clearly annunciated safe state, and connects to the aircraft through two galvanically-isolated channels: a receive-only INPUT tap and a dedicated isolated advisory-audio OUTPUT channel (COM3-style). This profile is intended to suit a NORSEE approval path emphasizing environmental qualification (e.g., RTCA/DO-160G) while avoiding complex software

assurance (e.g., RTCA/DO-178C). The isolated audio OUTPUT into the COM3-style channel is the most significant interface-risk item: the isolation design must demonstrate that a device fault cannot key, jam, load, or back-feed the panel's required channels, and that the injected advisory audio cannot mask or be mistaken for required ATC/aircraft audio. These questions must be addressed in the isolation design and safety assessment. (This posture is a design goal and roadmap; the prototype holds no approvals.)

9. Alternatives and scope

The disclosure contemplates numerous alternatives, including: other microcontroller/SoC families and other speech frameworks; other isolation means (optical or capacitive isolation, transformer-coupled digital); other removable media (e.g., other flash card formats, or two partitions/namespaces on physically distinct media); cryptographic signing/checksums of either card; additional fault conditions and additional annunciator states; an optional visual display in lieu of or in addition to the dark-cockpit annunciator; and use in fixed-wing, rotorcraft, simulator, and training contexts. Features described together may be claimed separately. The invention is not limited to the specific parts, values, formats, or aircraft examples named herein, which are illustrative.

10. Cryptographic card integrity and authentication

In a further aspect, the apparatus verifies the authenticity and integrity of each removable card before relying on its contents. Each card carries, in addition to its manifest and (for the data card) its read-aloud clips, a cryptographic integrity value — for example, a per-file and/or whole-manifest cryptographic hash (e.g., SHA-256), a keyed message authentication code (e.g., HMAC), or a digital signature produced with a private key under the control of the content author/installer. At boot, and as part of the validation described in Section 6, the apparatus recomputes the integrity value over the card contents and compares it against the stored value, and (for signatures) verifies the signature against a public key held in device firmware or in a write-protected region of the configuration card. A mismatch — indicating a corrupted, truncated, tampered-with, or unauthenticated card — is treated as a fault condition under the revert-to-unopened rule of Section 6: the apparatus refuses to present any checklist and announces a fault. This provides tamper-evidence and authenticity assurance, ensures that only content authored by an authorized party is presented, and detects silent media corruption that schema validation alone would not catch. The integrity scheme may be applied independently to the configuration card and the data card, and the configuration card may carry the public key(s) used to authenticate the data card so that authority to author content is delegated by the installation-controlled card.

11. Tamper-evident configuration-control and audit trail

In a further aspect, the configuration card maintains a per-installation, append-only provenance and audit record, and the data card maintains an append-only content-revision history. Each record comprises a sequence of entries, each entry recording an event (e.g., initial installation, configuration change, card re-imaging, content revision) together with metadata (shop, installer, date/time, work-order or revision identifier). Each entry is cryptographically linked to the prior entry — for example, each entry includes a hash computed over the prior entry (a hash chain) and/or a signature — such that deletion, reordering, or back-dating of any entry is detectable by recomputing the chain. Combined with the preferably write-protected/locked state of the configuration card after installation (Section 5), this yields a tamper-evident configuration-control record suitable for supporting the installation's airworthiness and configuration-management documentation: a maintainer or inspector can verify that the in-service configuration matches the recorded, authorized configuration and that the record has not been altered. The audit trail is advisory record-keeping and does not itself gate flight operations beyond the existing revert-to-unopened rule.

12. Multi-aircraft / fleet data card

In a further aspect, a single data card carries content for a plurality of aircraft (a fleet), each as a separate, independently-identified content set (e.g., one folder per tail/type, each with its own manifest, trigger/advance vocabulary, and read-aloud clips). The configuration card names the active aircraft identifier for the particular installation, and the cross-card consistency check of Section 6 selects, from among the plural content sets on the data card, the single set whose identifier matches the configuration card. If no content set on the data card matches the configuration card's aircraft identifier — or if more than one ambiguously matches — the apparatus treats this as a cross-card mismatch fault under the revert-to-unopened rule and presents nothing. This allows one physical data card to be provisioned across a fleet or across multiple aircraft types while the installation-controlled configuration card deterministically binds a given installation to exactly one content set, preserving the installation-authority/content-authority separation of Section 5.

13. Active output-injection safeguards and non-interference enforcement

In a further aspect, the apparatus actively enforces non-interference of its advisory OUTPUT channel (Section 2a) with required aircraft communications, by one or more of the following safeguards, alone or in combination:

- Keying-aware output inhibit. A detector senses a transmit/keying condition on a required COM radio or the intercom (for example, by monitoring a PTT-sense input, a keyline/relay state, or detecting transmit-side audio), and the apparatus automatically inhibits or mutes its OUTPUT stage for the duration of, and for a configurable hangover after, the keying event, so advisory audio is never injected while the crew is transmitting.
- Mute-on-fault. Upon any detected fault (including the faults of Section 6, a codec/DAC fault, or loss of a valid checklist store), the OUTPUT stage is driven to a defined muted/quiescent state rather than an indeterminate one.
- Output isolation self-test. At power-up (and optionally periodically), the apparatus performs a self-test of the OUTPUT isolation barrier and line driver — for example, by exercising a known test tone through the output path with the panel injection point disabled, or by sensing the barrier's continuity/leakage — and treats a failed self-test as a fault under the revert-to-unopened rule, so a degraded output stage cannot silently operate.
- Playback watchdog. A watchdog monitors the playback sequencer; if playback stalls, runs long, or repeats unexpectedly, the OUTPUT is muted and a fault is annunciated, preventing a stuck or runaway advisory output.

These safeguards make the non-interference property of the OUTPUT channel an actively enforced and testable behavior, not merely a passive consequence of the isolation transformer, and directly address the principal interface-risk item identified in Section 8.

14. No-mask audio arbitration

In a further aspect, the apparatus arbitrates its advisory read-aloud audio against required communications so that advisory audio cannot mask, overlap, or be mistaken for required ATC or aircraft audio. A detector (which may reuse the receive-only INPUT tap of Section 2 and/or the VAD of Section 3) senses the presence of required-radio/ATC or intercom audio. When such audio is present, the apparatus automatically ducks (reduces the level of), pauses, or defers the current advisory read-aloud item, and resumes or repeats it once the required audio clears (optionally after a configurable hangover). The

arbitration policy (duck vs. pause vs. defer, thresholds, and hangover) is configurable from removable media. This ensures the advisory channel yields to required communications and that a checklist item is fully and intelligibly delivered rather than being talked over.

15. Crew readback verification

In a further aspect, the apparatus advances through a checklist not merely upon a generic advance word but upon recognition of a correct item-specific spoken response (a "readback"). For one or more checklist items, the data card defines an expected response — for example, a target state or value such as "up," "on," "armed," or a spoken number — drawn from the bounded grammar. After reading the item, the apparatus awaits the expected response; upon recognizing the correct readback it advances, and upon recognizing an incorrect or out-of-grammar response it does not silently advance but instead re-prompts, repeats the item, or annunciates a mismatch as configured. Items may be configured to accept either a generic advance word (Section 4) or a specific readback, allowing a mixed checklist of acknowledge-only and verified-readback items. This raises the device from a read-and-advance reader toward a verifying challenge-and-response aid while remaining advisory and within the bounded, deterministic grammar.

16. Alternate isolation embodiments (claimed)

In a further aspect, the galvanic isolation barrier of the INPUT stage (Section 2), the OUTPUT stage (Section 2a), or both, is implemented by any of a plurality of interchangeable isolation means, including: a magnetic/transformer barrier (e.g., an audio isolation transformer); an optical isolation barrier (e.g., an opto-isolator or optical link carrying the audio or a digitized representation thereof); a capacitive isolation barrier (e.g., a capacitively-coupled digital isolator); or a transformer-coupled digital isolator carrying a digitized audio stream across the barrier. The two channels' barriers may be implemented by the same or different isolation means, and remain independent such that no conductive path is shared between the INPUT and OUTPUT interfaces. Reciting these alternatives expressly is intended to protect the dual-isolated-channel architecture against design-arounds that substitute one isolation technology for another.

17. Non-aircraft and training embodiments (claimed)

In a further aspect, the apparatus and method are embodied in non-aircraft and training contexts that likewise use an audio intercom or audio-distribution system, including: flight simulators and aviation training devices (where the "audio panel" is the simulator's audio system and no airworthiness approval is implicated); rotorcraft; and other crewed vehicles or control rooms employing an intercom or audio panel (e.g., maritime vessels, where the apparatus reads emergency or operating procedures aloud and advances on recognized crew speech via a receive-only isolated tap and a dedicated isolated advisory-audio output). In a training embodiment, the device may additionally log readback performance (Section 15) to the data or a separate card for instructor review. These embodiments practice the same dual-isolated-channel architecture, bounded-grammar offline recognition, dual-card configuration, and revert-to-unopened fail-safe described above, in a non-airworthiness-regulated setting.

Informal Claim-Style Statements (for drafting the later nonprovisional)

A provisional does not require claims; these statements are included to anchor scope and aid the nonprovisional. They are not formal claims.

1. An advisory aircraft checklist apparatus comprising: a receive-only, galvanically-isolated audio-input stage configured to derive crew speech from an INPUT tap of an aircraft audio panel, wherein the INPUT tap provides no capability to transmit on, key, mute, or back-feed the audio panel via that path; a dedicated, galvanically-isolated audio-output stage configured to inject checklist read-aloud audio into a separate, dedicated auxiliary channel of the audio panel (a "COM3-type" channel), the output stage including an isolation barrier such that a fault in the apparatus cannot key, jam, load, or back-feed the panel's other channels or required COM radios; wherein the INPUT stage and the OUTPUT stage are on separate galvanically-isolated paths with no shared conductive path therebetween; an offline microcontroller executing a bounded-grammar speech recognizer with a voice-activity detector; and a controller configured to play stored audio for each item of a checklist in sequence via the dedicated audio-output stage and to advance upon recognition of a spoken advance word.
2. The apparatus of statement 1, wherein the audio-input stage comprises an audio isolation transformer coupling a high-impedance analog tap of the audio panel to a codec, such that an internal fault or power loss cannot load down or back-feed the audio panel via the input path.
3. The apparatus of statement 1, wherein the audio-input stage comprises a buffered, receive-only digital feed to a codec, all clocks for which are generated locally with no data driven toward any aircraft bus, the analog and digital input paths being selectable per installation.
4. The apparatus of statement 1, wherein the speech recognizer operates in a voice-activated (VOX) mode gated by the voice-activity detector, with a push-to-talk input retained as a manual force-listen override.
5. The apparatus of statement 1, further comprising two physically separate removable-card slots receiving a configuration card defining an installation — including an aircraft identifier, a selected audio source, and voice-activation parameters — and a data card carrying aircraft-specific checklist content and read-aloud audio, whereby a new aircraft type is supported by a new data card without firmware change.
6. The apparatus of statement 5, wherein the configuration card is write-protected after installation and serves as a per-installation configuration-control record, separating installation authority from content authority.
7. The apparatus of statement 5, wherein the controller, before presenting any checklist, validates that both cards are present, readable, schema-valid, and that the aircraft identifier on the configuration card is consistent with the data card, and otherwise holds an empty checklist store and refuses to present any checklist.
8. The apparatus of statement 7, wherein refusal to present a checklist occurs upon any of: a missing or unreadable card; a malformed or schema-invalid manifest; a referenced audio clip being absent; or a cross-card aircraft-identifier mismatch — such that the apparatus never presents partial, stale, or mismatched checklist content.
9. The apparatus of statement 1, further comprising a dark-cockpit annunciator that shows nothing when selected-in and healthy, illuminates an amber fault legend on any fault, shows a white off-status and inhibits the fault legend when selected-out, and performs a power-up lamp test.
10. A method of advisory checklist reading comprising: tapping an aircraft audio panel through a receive-only, galvanically-isolated INPUT interface; recognizing, offline on a microcontroller using a bounded grammar gated by a voice-activity detector, a spoken checklist-invocation phrase in the tapped audio; playing an audio clip for each checklist item in sequence through a separate, dedicated,

galvanically-isolated OUTPUT channel of the audio panel, such that crew hear checklist items in-headset; advancing upon recognizing a spoken advance word; and, prior to presenting any checklist, requiring a configuration card and a data card to be present, valid, and mutually consistent as to aircraft identity, and otherwise entering an annunciated fault state without presenting any checklist content.

11. The method of statement 10, wherein supporting an additional aircraft type comprises installing a data card carrying that aircraft's checklist content and audio without modifying device firmware.

12. The apparatus of statement 1, wherein the microcontroller has no network, wireless, or cloud connectivity and the speech recognition is deterministic and limited to a fixed enumerated vocabulary loaded from removable media.

13. The apparatus of statement 1, wherein the audio-input stage and the audio-output stage each comprise a galvanic isolation barrier (e.g., an audio isolation transformer) on their respective audio lines, the two barriers being independent such that the apparatus has no conductive path shared between the aircraft audio panel's input tap connection and the apparatus's COM3-type output connection — a dual-isolated-channel architecture enabling the apparatus to receive crew audio for recognition and to inject advisory audio in-headset while maintaining electrical separation on both paths.

14. The apparatus of statement 13, wherein the audio-output stage isolation barrier is a line-level isolation transformer interposed between an I2S DAC/line driver output of the apparatus and the audio panel's COM3-type channel input, such that a fault, short-circuit, power loss, or output-stage failure in the apparatus cannot assert a voltage, current, or keying signal on the audio panel's required COM or intercom channels.

15. The apparatus of statement 1, wherein at least one of the configuration card and the data card carries a cryptographic integrity value over its contents — selected from a cryptographic hash, a keyed message authentication code, and a digital signature — and the controller, before presenting any checklist, recomputes and verifies the integrity value and treats a mismatch as a fault that causes the apparatus to refuse to present any checklist.

16. The apparatus of statement 15, wherein the integrity value is a digital signature verified against a public key, and the configuration card carries the public key used to authenticate the data card, whereby authority to author content is cryptographically delegated by the installation-controlled configuration card.

17. The apparatus of statement 5, wherein at least one card maintains an append-only, cryptographically-chained provenance/audit record in which each entry is linked to a prior entry by a cryptographic hash, such that deletion, reordering, or back-dating of an entry is detectable, providing a tamper-evident configuration-control record.

18. The apparatus of statement 5, wherein the data card carries content for a plurality of aircraft as plural independently-identified content sets, and the controller selects, as the active content set, the single set whose aircraft identifier matches the configuration card, and refuses to present any checklist if no content set matches or if the match is ambiguous.

19. The apparatus of statement 1, further comprising a non-interference safeguard for the audio-output stage comprising at least one of: (a) an output inhibit responsive to a detected transmit/keying condition on a required radio or intercom that mutes the output during and for a hangover after the keying condition; (b) a mute-on-fault that drives the output stage to a defined quiescent state upon any

detected fault; (c) a power-up self-test of the output isolation barrier that, on failure, prevents presentation of any checklist; and (d) a watchdog that mutes the output and annunciates a fault if the playback sequencer stalls.

20. The apparatus of statement 19, wherein the output inhibit is responsive to a push-to-talk/keyline sense input indicating that a required COM radio is transmitting, such that advisory audio is never injected into the audio panel while the crew is transmitting on a required radio.

21. The apparatus of statement 1, further comprising an audio-arbitration subsystem that detects the presence of required-radio or intercom audio and, in response, automatically ducks, pauses, or defers the advisory read-aloud audio and thereafter resumes or repeats it, such that advisory audio does not mask or overlap required communications.

22. The apparatus of statement 1, wherein, for at least one checklist item, the data card defines an expected item-specific spoken response within the bounded grammar, and the controller advances only upon recognizing the correct expected response and, upon recognizing an incorrect or out-of-grammar response, re-prompts or repeats the item rather than advancing.

23. The apparatus of statement 22, wherein a checklist comprises a mixture of items configured to advance on a generic advance word and items configured to advance only on a correct item-specific readback.

24. The apparatus of statement 1, wherein the galvanic isolation barrier of at least one of the input stage and the output stage comprises an isolation means selected from: a magnetic/transformer barrier, an optical isolation barrier, a capacitive isolation barrier, and a transformer-coupled digital isolator, the input and output barriers being independent with no shared conductive path therebetween.

25. A training or simulator embodiment of the apparatus of statement 1, wherein the aircraft audio panel is a simulator or training-device audio system, the apparatus reading procedure items aloud and advancing on recognized crew speech via a receive-only isolated tap and a dedicated isolated advisory-audio output, optionally logging readback performance for instructor review.

26. A non-aircraft vehicle or control-room embodiment of the apparatus of statement 1, wherein the audio panel is an intercom or audio-distribution system of a crewed vehicle or facility, the apparatus reading operating or emergency procedures aloud and advancing on recognized crew speech.

27. The method of statement 10, further comprising verifying a cryptographic integrity value over at least one card before presenting any checklist and entering an annunciated fault state on a mismatch.

28. The method of statement 10, further comprising inhibiting or muting the advisory audio output upon detecting that a required radio is transmitting, and ducking, pausing, or deferring the advisory audio upon detecting required-radio or intercom audio, so as not to interfere with or mask required communications.

29. The method of statement 10, wherein advancing comprises recognizing a correct item-specific readback within the bounded grammar for at least one item, and re-prompting rather than advancing upon an incorrect readback.

30. The method of statement 10, wherein a single data card carries content for a plurality of aircraft and the method selects the active content set as the one whose aircraft identifier matches the configuration card, entering an annunciated fault state if none matches.

Abstract

An offline, standalone advisory device reads aircraft checklist items aloud and advances on recognized crew speech. The device connects to an aircraft audio panel through two electrically separate, galvanically-isolated channels: (1) a receive-only INPUT tap (isolation-transformer-coupled analog or buffered-receive-only digital feed, selectable per installation) used as the speech-recognition audio source, such that the device cannot back-feed the panel via the input path; and (2) a dedicated isolated OUTPUT stage that injects checklist read-aloud audio into a separate auxiliary audio-panel channel ("COM3-type"), so crew hear checklist items in-headset, with the isolation barrier ensuring a device fault cannot key, jam, or back-feed the panel's required channels. The two channels are on separate paths with independent isolation barriers. A microcontroller runs an entirely on-device, deterministic, bounded-grammar speech recognizer with a voice-activity detector enabling hands-free (VOX) operation, with push-to-talk retained as a manual override. The device is rendered aircraft-specific by two removable cards — a write-protected configuration card (installation parameters) and a data card (checklist content and audio) — and supports new aircraft types without firmware change. Before presenting any checklist, the device requires both cards to be present, valid, and consistent as to aircraft identity; otherwise it presents nothing and enters a dark-cockpit-annunciated fault state, never showing partial or stale content.

Prosecution notes (internal — remove before filing)

- Closest active art to distinguish: US7289890B2 (Boeing Voice Checklist System) and US9550578B2 (Honeywell voice commands onboard aircraft). Neither claims the receive-only isolated audio-panel tap, the dedicated isolated output into an audio-panel aux channel, the dual-card config/content split with cross-card consistency, or the revert-to-unopened fail-safe.
- Closest expired art (free to practice, but citable): US7912592B2 (Garmin ASR for aircraft) covers the basic speak-name → TTS → "Check"-advance loop. Distinguish on offline/microcontroller, audio-panel tap, VOX-primary, dual-card, fail-safe, and dual-isolated-channel aspects.
- Prior-art search scope re-scoping required (important). The original provisional was built around the absolute claim that the device "cannot transmit/inject into/back-feed the audio panel." That absolute claim is now re-scoped: the INPUT tap remains receive-only and cannot back-feed, but the OUTPUT stage is a new, intentional audio injection into a dedicated aux channel. The prior-art search was conducted under the old scope and did not search for prior art on injecting advisory audio into an audio panel / intercom audio injection (e.g., audio injection for cockpit entertainment, audio panel aux-channel injection, COM3-style aux audio). The attorney must conduct an additional prior-art search covering intercom/audio-panel audio injection before relying on the white-space conclusions for the output feature. The receive-tap white space remains valid; the output injection white space is unverified.
- Potential new novelty — dual-isolated-channel architecture. The combination of a receive-only isolated INPUT tap AND a separately isolated dedicated advisory-audio OUTPUT channel (both galvanically isolated, electrically separate, on independent barriers) may itself be a novel combination not found in the prior art search. Claim statements 13 and 14 are directed to this architecture. Attorney should evaluate this as an independent claim anchor.
- Strongest independent claims likely lie in: (a) the dual-isolated-channel architecture (new claim 13/14); (b) the receive-only isolated INPUT tap as ASR source (Feature 2, carry forward); (c) dual-card cross-consistency (Feature 4); (d) revert-to-unopened (Feature 5). Treat Feature 1 (the loop) as background and claim it only in combination.

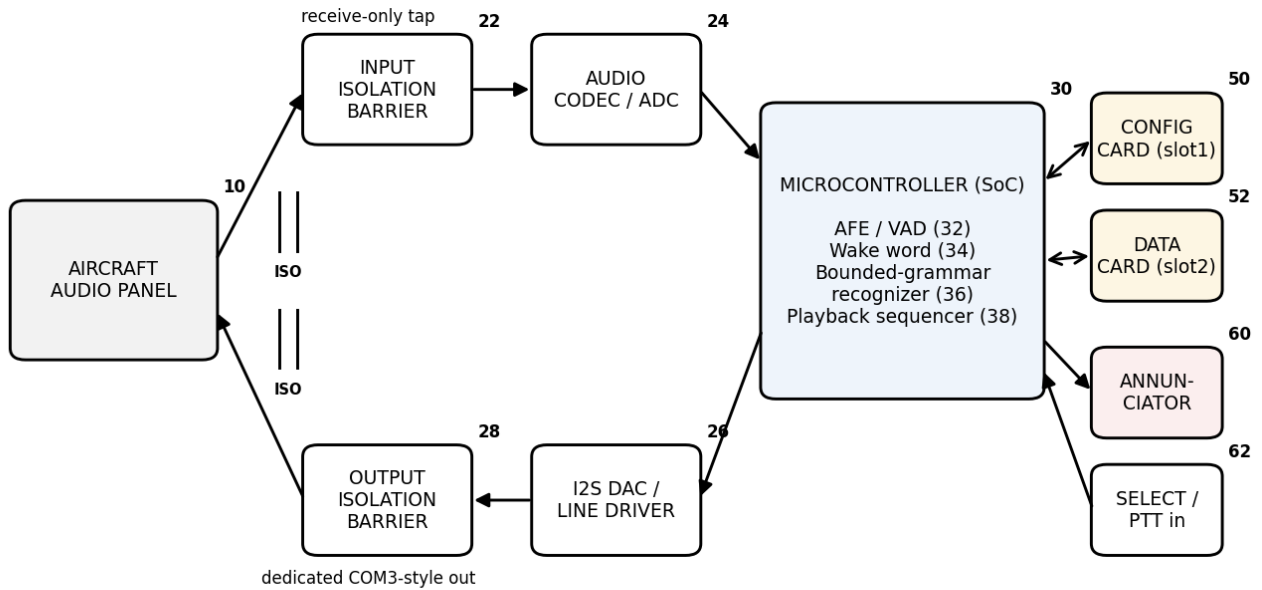
- Architecture change note: the prior design used an onboard speaker/amplifier ("not fed back to the panel"). That claim was the core novelty framing. The new design REMOVES the onboard speaker and instead USES a dedicated isolated output channel into the audio panel. This is a material change that (i) improves crew ergonomics (in-headset delivery), (ii) adds the output as a new potential novelty, and (iii) requires narrowing the absolute "cannot inject into the panel" language to apply only to the INPUT tap. Prior claim 1 (old) asserted the device had "no capability to inject into the panel" — that language now applies only to the INPUT tap path; the new OUTPUT stage is an intentional, isolated, advisory injection.
- Add formal figures before nonprovisional. Update FIG. 1 to show the dual-channel block diagram (audio-panel receive tap AND isolated COM3 output stage). The provisional can be filed with informal drawings.
- New feature sections 10–17 / claims 15–30 added (June 2026). The following were added to broaden the disclosure and seed continuation/nonprovisional claims: (10) cryptographic card integrity/authentication; (11) tamper-evident audit trail; (12) multi-aircraft/fleet data card; (13) active output-injection safeguards (keying-aware inhibit, mute-on-fault, output-isolation self-test, playback watchdog); (14) no-mask audio arbitration; (15) crew readback verification; (16) alternate isolation embodiments (optical/capacitive/digital), claimed; (17) non-aircraft/training embodiments, claimed. None of these new features were within the scope of the original prior-art search. Adding them to the provisional is appropriate (broad disclosure supports later claims and they are protected as of this filing date), BUT each feature relied upon for novelty in the nonprovisional requires its own prior-art / patentability search before claims are drafted around it. In particular: (a) output-injection safeguards and no-mask arbitration overlap the same audio-injection white-space gap already flagged for the output channel — search audio-panel/intercom audio injection, COM-keying-aware muting, and audio ducking/arbitration in cockpit audio; (b) crew readback verification — search aviation challenge-and-response / readback-verification voice systems (note Garmin/Honeywell references advance on a generic word, not a verified readback, but a dedicated search is required); (c) cryptographic card integrity and hash-chained audit — search signed-removable-media / secure-configuration and tamper-evident maintenance-log art (these are common concepts outside aviation; novelty likely lies only in the avionics-checklist combination, so claim them in combination); (d) multi-aircraft data card — search multi-config removable media for avionics. Treat the new features as combination-claim material layered on the lead novelties (dual-isolated-channel, receive-only tap, dual-card cross-consistency, revert-to-unopened), not as standalone broad claims.
- Figures for new features: informal FIG. 8–14 added to the drawings list to support sections 10–17; these may be filed as informal drawings with the provisional and formalized before the nonprovisional.

DRAWINGS

FIG. 1 – FIG. 14. Informal drawings (acceptable for a provisional application under 37 CFR 1.81).

FIG. 1

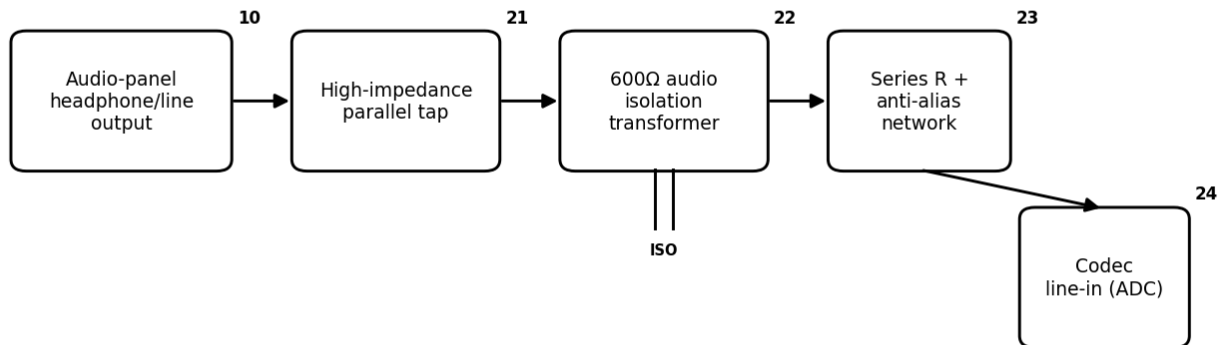
System Block Diagram — Dual Galvanically-Isolated Channels



Two electrically-separate, galvanically-isolated channels: receive-only INPUT tap (22/24) and dedicated isolated OUTPUT (26/28).

FIG. 2

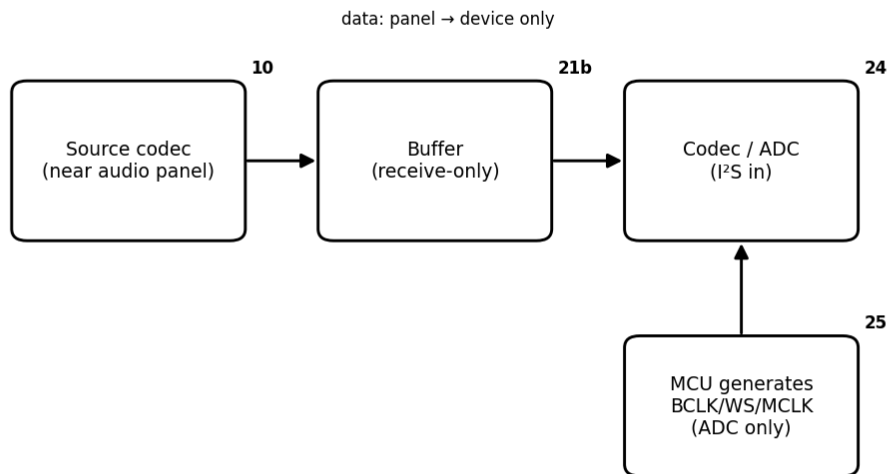
Analog Audio-Input Embodiment (Receive-Only, Isolated)



High-impedance parallel tap loads the panel negligibly; transformer (22) blocks any back-feed via the input path.

FIG. 3

Digital Audio-Input Embodiment (Buffered, Receive-Only)



All clocks generated locally for the ADC; no data line is driven back toward any aircraft bus.

FIG. 4

Dual-Card Configuration Architecture & Cross-Card Check

Installation authority (config) is separated from content authority (data).

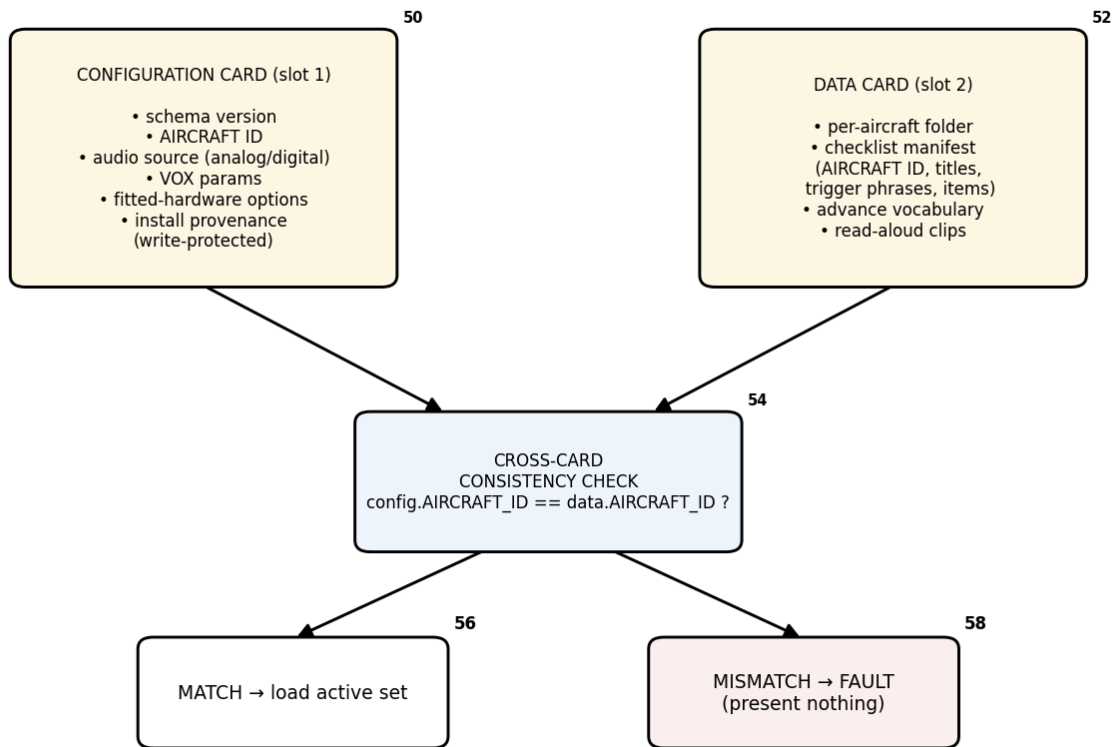


FIG. 5

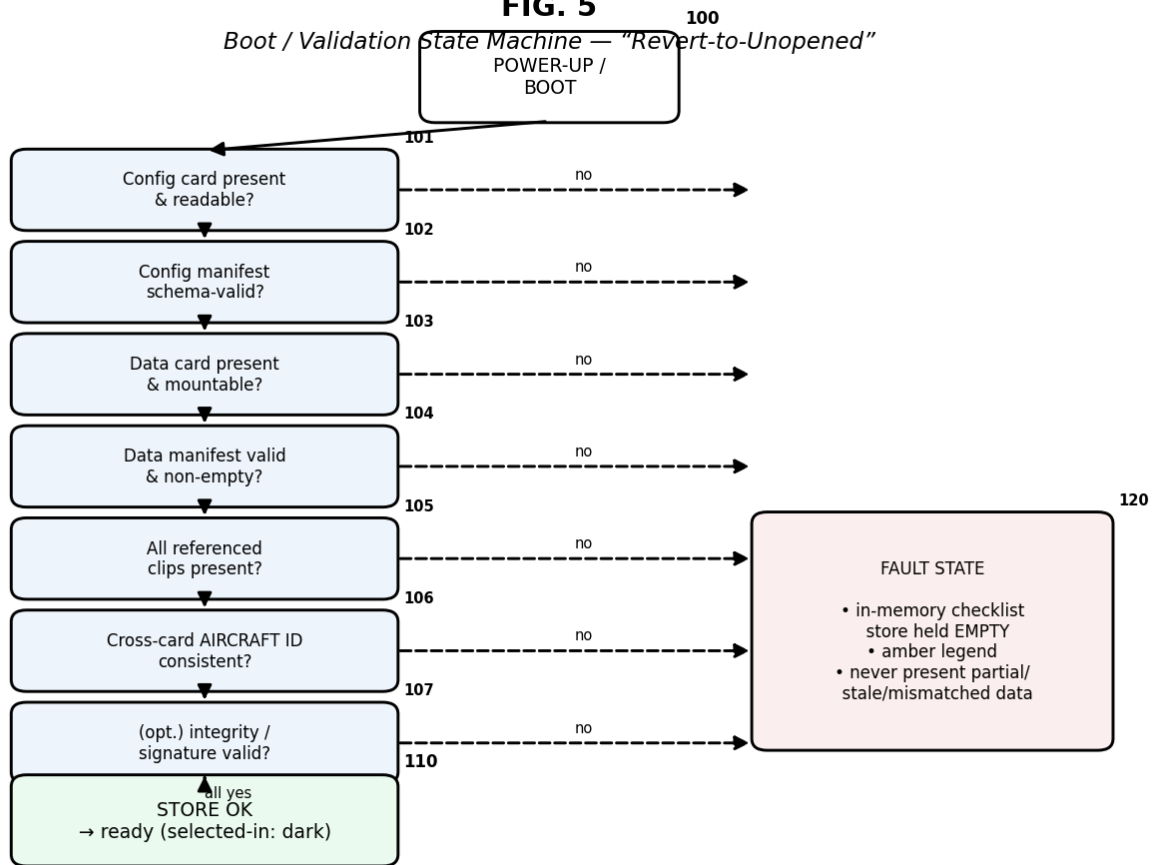


FIG. 6

Dark-Cockpit Annunciator States & Power-Up Lamp Test

CONDITION	ANNUNCIATOR
Selected-IN, healthy	DARK (nothing shown)
Any fault	AMBER 'FAULT' legend
Selected-OUT	WHITE 'OFF'; fault inhibited
Power-up lamp test	All legends ON briefly

A power-up lamp test verifies the indicator so a failed lamp cannot masquerade as a healthy/dark state.

FIG. 7

Interaction Flow — Hands-Free (VOX) with PTT Override

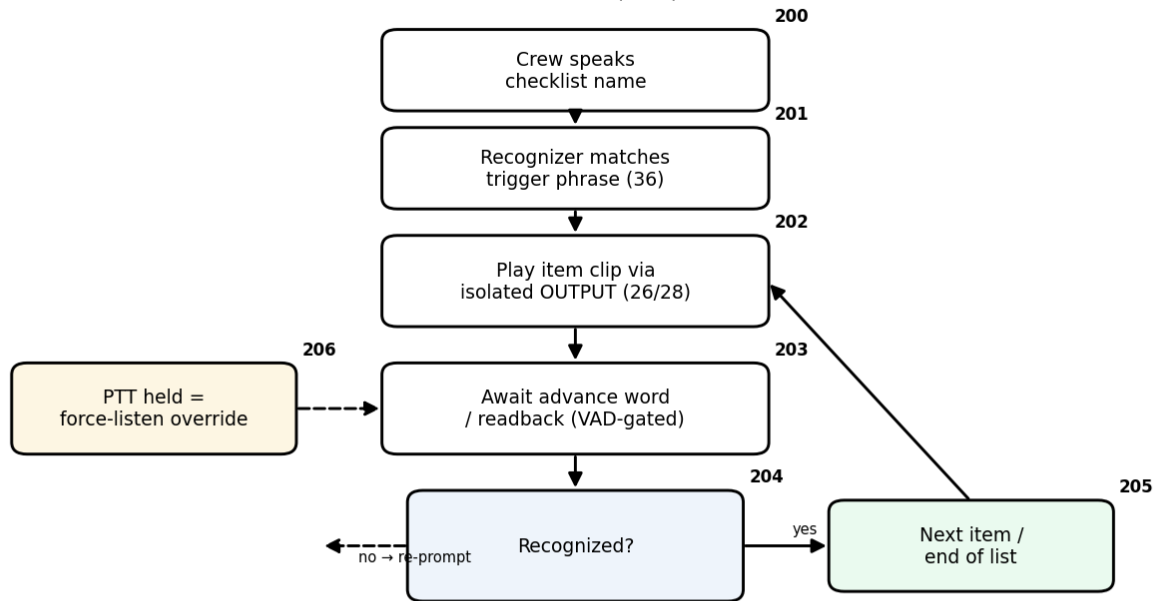
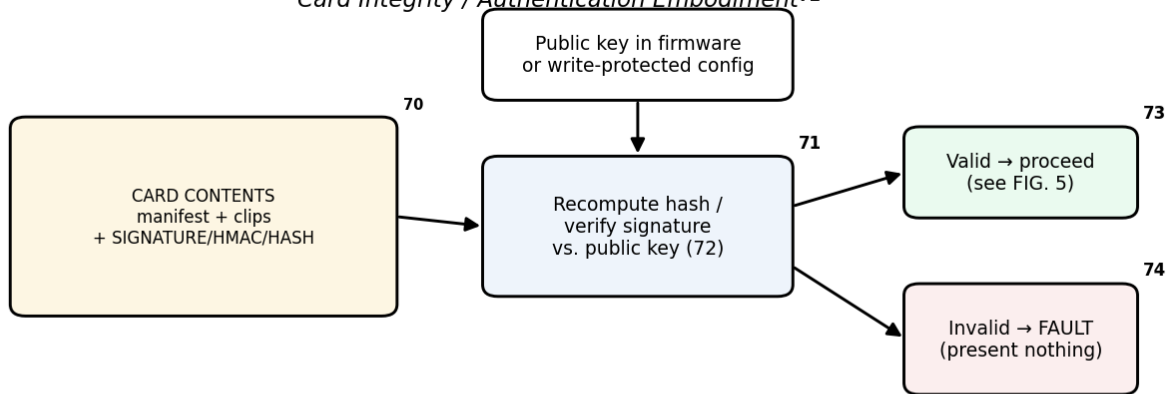


FIG. 8

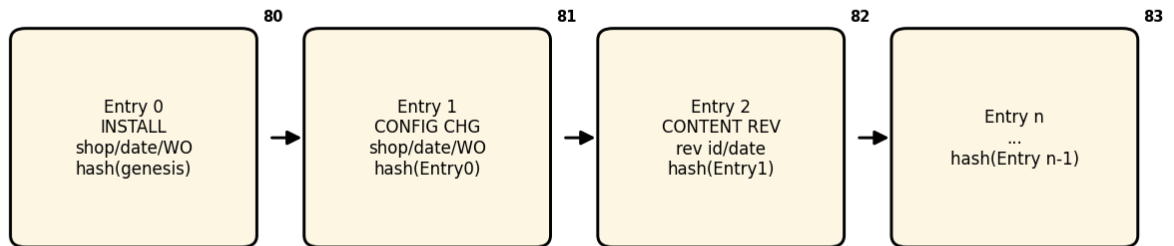
Card Integrity / Authentication Embodiment⁷²



Tampered, corrupted, or unauthenticated cards are rejected under the revert-to-unopened rule.

FIG. 9

Tamper-Evident Configuration / Audit Log (Hash-Chained)



Each entry includes a hash over the prior entry; deletion, reordering, or back-dating is detectable by recomputing the chain.

FIG. 10

Multi-Aircraft (Fleet) Data Card with Active-Set Selection

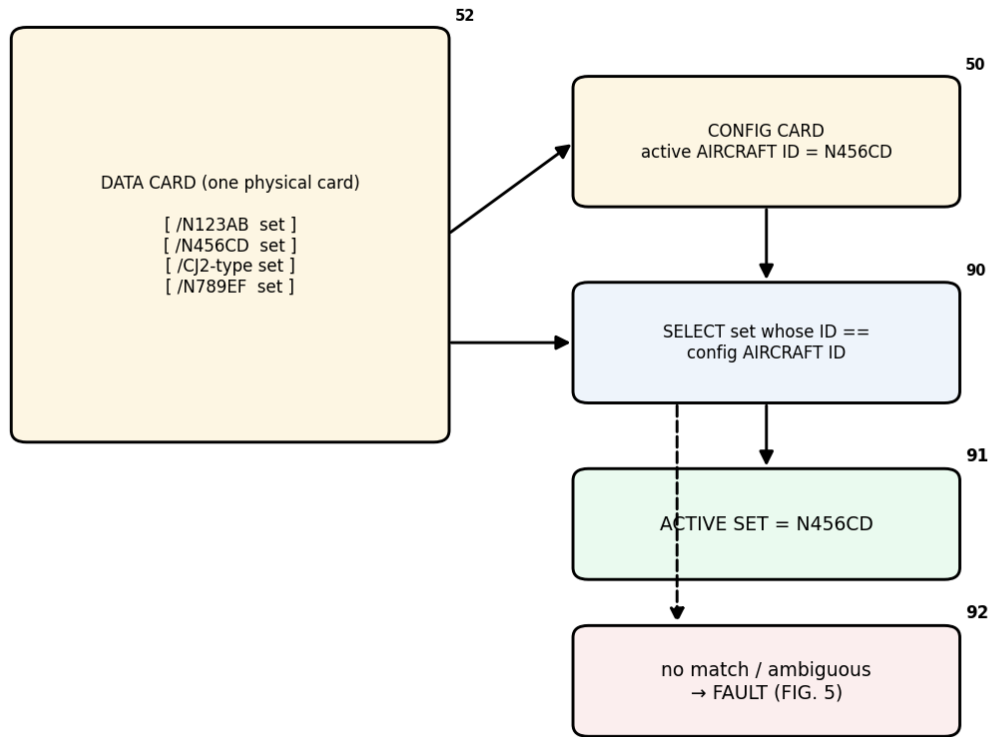
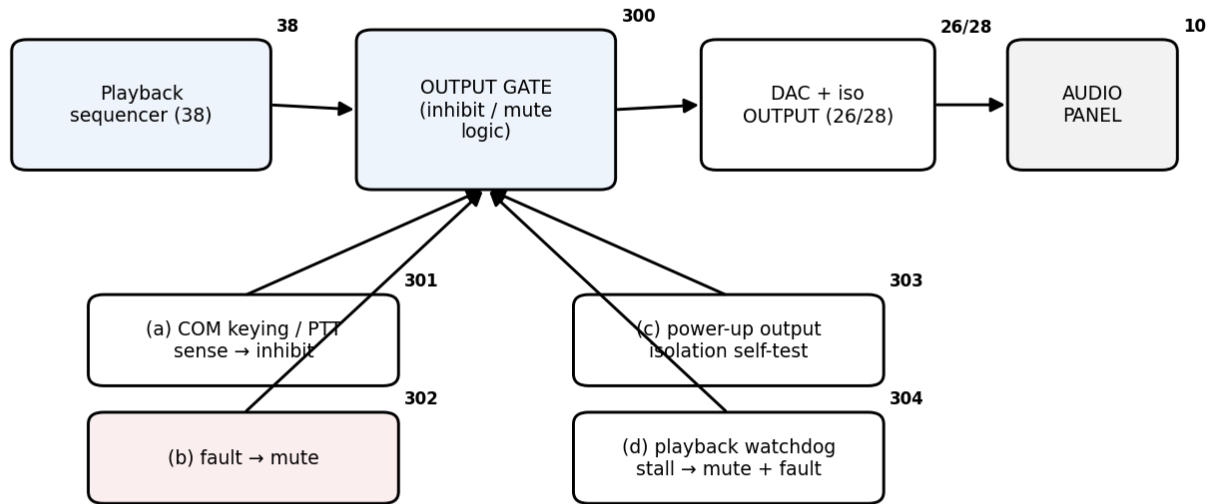


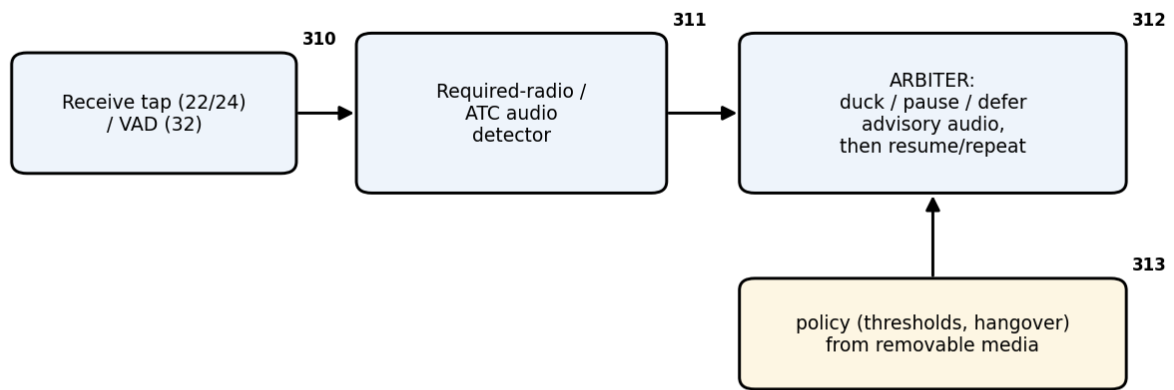
FIG. 11
Active Output-Injection Safeguards (Non-Interference)



Non-interference is an actively enforced, testable behavior — not merely a passive consequence of the transformer.

FIG. 12

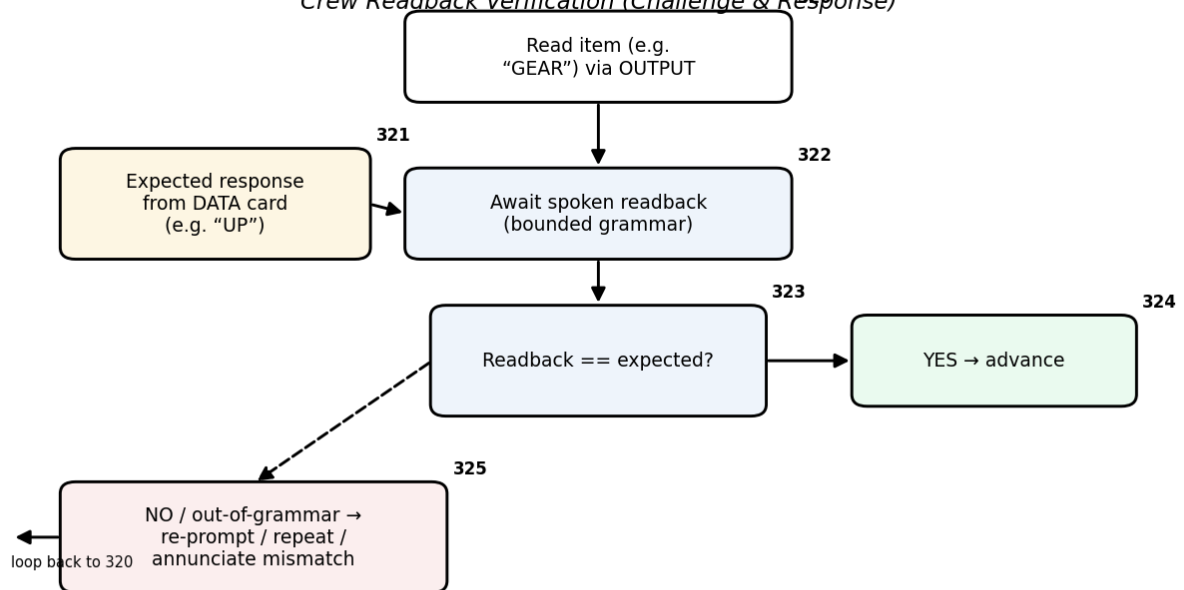
"No-Mask" Audio Arbitration vs. Required Communications



Advisory read-aloud yields to required communications so a checklist item is delivered intelligibly and never talks over ATC.

FIG. 13

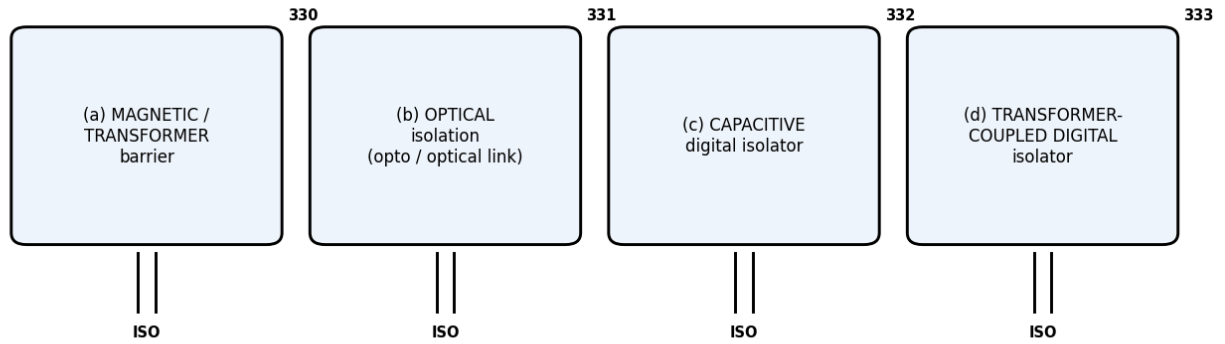
Crew Readback Verification (Challenge & Response)



Items may be configured to accept a generic advance word OR a specific verified readback.

FIG. 14

Interchangeable Isolation-Barrier Embodiments (Either Channel)



Input and output barriers remain INDEPENDENT — no shared conductive path — regardless of which isolation means is used.

Reciting alternatives expressly protects the dual-isolated-channel architecture against substitute-technology design-arounds.